$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$

$\Delta U = -1406$ kJ

$\Delta n = n_{p(g)} - n_{r(g)}$

$\Delta n = 2 - 4 = -2$

$\Delta H = \Delta U + RT\Delta n_g$

$\Delta H = -1406 + (8.314 \times 10^{-3} \times 300 \times (-2)$

$\Delta H = -1410.9$ kJ

Evaluate Yourself - 3

From the following data at constant volume for combustion of benzene, calculate the heat of this reaction at constant pressure condition.

$C_6H_6(l) + 7½\ O_2(g) \rightarrow$

$6CO_2(g) + 3\ H_2O(l)$

ΔU at 25^0 C = -3268.12 kJ

Applications of the heat of combustion:

(1) Calculation of heat of formation: Since the heat of combustion of organic compounds can be determined with considerable ease, they are employed to calculate the heat of formation of other compounds.

For example let us calculate the standard enthalpy of formation ΔH_f^0 of CH_4 from the values of enthalpy of combustion for H_2, C(graphite) and CH_4 which are - 285.8, - 393.5, and -890.4 kJ mol^{-1} respectively.

Let us interpret the information about enthalpy of formation by writing out the equations. It is important to note that the standard enthalpy of formation of pure elemental gases and elements is assumed to be zero under standard conditions. Thermochemical equation for the formation of methane from its constituent elements is,

$C_{(graphite)} + 2H_2(g) \rightarrow CH_4(g)$

$\Delta H_f^0 = X$ kJ mol^{-1}--- (i)

Thermo chemical equations for the combustion of given substances are,

$H_2(g) + \frac{1}{2} O_2 \rightarrow H_2O(l)$

$\Delta H^0 = -285.8$ kJ mol^{-1}--- (ii)

$C_{(graphite)} + O_2 \rightarrow CO_2$

$\Delta H^0 = -393.5$ kJ mol^{-1}--- (iii)

$CH_4(g) + 2\ O_2 \rightarrow CO_2(g) + 2H_2O(l)$

$\Delta H^0 = -890.4$ kJ mol^{-1}--- (iv)

Since methane is in the product side of the required equation (i), we have to reverse the equation (iv)

$CO_2(g) + 2\ H_2O(l) \rightarrow CH_4(g) + 2\ O_2$

$\Delta H^0 = +890.4$ kJ mol^{-1}--- (v)

In order to get equation (i) from the remaining,

(i) = [(ii) × 2] + (iii) + (v)

$X = [(-285.8) \times 2] + [-393.5] + [+890.4]$

$= -74.7$ kJ

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